

Lifelong Opportunity Guide 83 — Network Support Technician

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Language: English

Primary U.S. benchmark: O*NET-SOC 15-1231.00 — Computer Network Support Specialists

Canada comparison: NOC 22220 — Computer network and web technicians

Colombia comparison: CUOC 35130 — Técnicos en redes y tecnologías de la información

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What this career is

A network support technician helps keep data networks available, usable, secure and supportable. The work can include troubleshooting LAN, WAN, wireless, VPN, cloud and hybrid connectivity; checking switches, routers, access points, firewalls and related services; reviewing logs and monitoring data; documenting incidents and changes; supporting users; backing up configurations; and escalating faults or security concerns that exceed assigned authority.

This guide uses **O*NET-SOC 15-1231.00 — Computer Network Support Specialists** as its primary United States benchmark. Canada maps closely to **NOC 22220 — Computer network and web technicians**. Colombia has a direct occupational match in **CUOC 35130 — Técnicos en redes y tecnologías de la información**.

Network support is not the same as network architecture, penetration testing, security engineering or unrestricted administrator authority. A technician may diagnose and correct many faults, but production access, configuration changes and security actions must stay within employer authorization and change-control rules.

Why this can be a strong opportunity

Organizations still depend on reliable connectivity across offices, data centers, cloud services, remote users, wireless networks and internet-facing systems. Even when routine monitoring and configuration become more automated, employers still need people who can interpret symptoms, isolate faults, communicate impact, validate changes, preserve evidence and know when to escalate.

The strongest long-term strategy is not to remain limited to repetitive scripted tasks. Build toward deeper networking, systems, cloud, automation, observability and cybersecurity capability.

A practical progression can look like:

help desk or field support → network support technician → network administrator / systems administrator / cloud support → network engineer, cloud engineer, security operations or infrastructure specialist.

Actual progression depends on experience, education, employer structure and jurisdiction.

Network support technician, network administrator and network engineer are not identical roles

Network support technician

Often focuses on:

- user and site connectivity incidents;
- network-device status and first-line diagnostics;
- cable, port, VLAN, IP, DNS, DHCP and wireless issues;
- monitoring alerts;
- approved configuration tasks;
- escalation and documentation;
- remote support and coordination with carriers/vendors.

Network administrator

May have broader responsibility for:

- ongoing network administration;
- account/access controls;
- backup/recovery;
- server/network services;
- patching and lifecycle management;
- more extensive change authority.

Network engineer or architect

May design and implement larger-scale routing, switching, wireless, cloud, WAN, segmentation, resilience and security architectures. These roles normally require deeper engineering judgment and broader change authority.

Security roles

Security analysts and engineers may investigate threats, tune controls, perform authorized testing or design security architecture. Network support staff should recognize and escalate security indicators but must not perform unauthorized scanning, exploitation or evidence destruction.

Professional authority boundaries

A network-support title does **not** automatically authorize a worker to:

- bypass access controls to troubleshoot;
- use another person's privileged account;
- disable a firewall, endpoint control, MFA or monitoring system because it appears inconvenient;
- scan or probe systems outside assigned authorization;
- perform penetration testing without explicit written scope;
- make production routing, switching, wireless, DNS or firewall changes outside approved change procedures;
- copy passwords, keys, tokens or protected configurations into public notes or unapproved AI services;
- erase logs or evidence after a suspected breach;
- claim a service is restored before validating it;
- make architecture, regulatory or legal decisions beyond assigned authority;
- claim certifications or credentials not actually held.

When a problem requires broader privileges, security investigation, emergency change authority or architecture redesign, escalate through the approved process.

A disciplined troubleshooting model

Strong network troubleshooting is evidence-driven. A practical sequence is:

1. define the user, service, site and business impact;
2. establish what changed and when;
3. verify whether the problem is isolated or widespread;
4. identify the expected path and dependencies;
5. collect evidence before changing anything;
6. test the lowest-risk hypotheses first;
7. make only approved, reversible changes;
8. validate service from the user's perspective;
9. record what was observed, changed and confirmed;
10. escalate unresolved or security-sensitive findings.

Avoid random command sequences. A command is useful only when you know what question it is intended to answer.

The OSI and TCP/IP models as troubleshooting tools

Models are useful when they help isolate a fault.

A simplified progression:

- **physical/link:** power, cable, optics, radio, interface state, errors;
- **data link:** MAC learning, VLAN membership, trunking, spanning tree;
- **network:** IPv4/IPv6 addressing, subnet, gateway, routing;
- **transport:** TCP/UDP reachability and ports;
- **application/service:** DNS, authentication, web, messaging, file or business application behavior.

Do not assume the problem is "the network" merely because an application is unavailable.

IP addressing and subnetting

A network technician should understand:

- IPv4 addresses and subnet masks/prefixes;
- private versus public address space;
- default gateways;
- network/broadcast concepts for IPv4;
- CIDR notation;
- IPv6 address and prefix basics;
- static versus dynamic addressing;
- duplicate addresses;
- address pools and exhaustion;
- how incorrect subnetting creates reachability problems.

Subnetting skill is especially useful when reading routing tables, DHCP scopes, ACLs and network diagrams.

DHCP, DNS and name resolution

DHCP

DHCP commonly provides clients with:

- IP address;
- subnet/prefix information;
- default gateway;
- DNS server information;

- lease timing and other options.

Troubleshooting may include checking lease state, scope availability, relay/helper behavior, VLAN placement and whether the client received expected options.

DNS

DNS problems can look like network failures. Verify:

- whether the host can reach the DNS server;
- whether the expected record exists;
- whether the client is using the correct resolver;
- whether cached/stale information is involved;
- whether the problem affects one name, one zone or all name resolution.

Do not make DNS changes without confirming authority and downstream impact.

Switching, VLANs and trunks

Useful switching concepts include:

- access ports;
- VLAN membership;
- trunk links;
- tagged/untagged traffic concepts;
- MAC address tables;
- loop prevention and spanning-tree concepts;
- port errors and negotiation;
- link aggregation concepts;
- switch management and configuration backup.

A common support error is to change a port or VLAN before confirming the intended design. Verify documentation, device, interface and endpoint identity first.

Routing and gateways

Useful routing concepts include:

- directly connected networks;
- static routes;
- default route;
- routing tables;
- next hop;

- administrative preference/metrics concepts;
- dynamic routing awareness;
- route asymmetry;
- network address translation concepts.

O*NET employer-posting data specifically identifies **Border Gateway Protocol (BGP)** among current technology signals. Entry-level technicians do not need to design internet-scale BGP, but they should understand that dynamic routing changes require careful authorization and validation.

Wireless networking

Wireless support can involve:

- SSID and authentication;
- access-point reachability;
- signal strength and interference;
- client compatibility;
- channel/congestion concepts;
- roaming;
- guest versus internal segmentation;
- captive portals;
- enterprise authentication;
- approved security settings.

Do not weaken wireless encryption or authentication merely to make a device connect.

VPN and remote connectivity

Remote-access problems may involve:

- internet reachability;
- DNS;
- identity/MFA;
- client configuration;
- certificate or credential state;
- split-tunnel/full-tunnel policy;
- routes;
- endpoint posture;
- firewall policy;
- licensing/capacity;
- service outage.

Respect privacy during remote sessions. Explain what you are doing, use approved tools, limit access to the task, and close the session when work is complete.

Network performance and availability

Useful measurements include:

- latency;
- packet loss;
- jitter;
- throughput;
- bandwidth utilization;
- interface errors/discards;
- retransmissions;
- availability and outage duration.

A slow application may be caused by the network, server, client, storage, authentication, DNS, application code or an external dependency. Correlate evidence before assigning cause.

Monitoring, logs and observability

Network support may use:

- interface/device monitoring;
- syslog/event logs;
- SNMP or other telemetry;
- flow data;
- packet captures where authorized;
- performance dashboards;
- alerting systems;
- configuration/history systems;
- ticket/event correlation.

Preserve time context and source identity. If a security incident is suspected, follow evidence-preservation and escalation procedures rather than deleting or "cleaning up" logs.

Cloud and hybrid networking

Modern support increasingly touches:

- virtual networks/VPC/VNet concepts;
- subnets and route tables;
- security groups/firewall rules;
- VPN/direct-connect concepts;

- load balancers;
- DNS;
- identity and access;
- cloud monitoring;
- hybrid connectivity between cloud and on-premises networks.

O*NET current postings include Microsoft Azure among technology signals. Learn transferable cloud-network concepts rather than assuming a single provider is universal.

Configuration control, backup and recovery

Before an approved production change, know:

- what is being changed;
- why;
- who approved it;
- dependencies;
- validation steps;
- rollback method;
- maintenance window if applicable;
- how configuration/evidence will be recorded.

Keep required backups and follow retention/security rules. A backup containing credentials or sensitive topology must be protected appropriately.

Change management

Good change records include:

- requested change;
- affected devices/services;
- risk/impact;
- approval;
- implementation plan;
- test/validation plan;
- rollback plan;
- actual result;
- incident/problem link if relevant.

Emergency changes may use a faster process, but "emergency" does not mean undocumented or unlimited authority.

Ticket and documentation quality

A strong network ticket should make the work reproducible. Capture:

- who/what is affected;
- location/site;
- device/service identifiers;
- symptoms and timestamps;
- business impact;
- diagnostics performed;
- evidence/results;
- changes made;
- validation;
- escalation/next action.

Avoid vague closures such as "fixed network." Explain what was wrong and what verified restoration.

Cybersecurity in network support

O*NET explicitly includes configuring security settings/access permissions and analyzing network-security breaches or attempted breaches among occupation tasks.

Support-level cybersecurity habits include:

- least privilege;
- named accounts;
- MFA where required;
- secure credential handling;
- approved configuration changes;
- patch/update procedures;
- phishing/social-engineering awareness;
- secure remote support;
- protection of network diagrams/configurations;
- escalation of suspicious traffic or access;
- preserving logs and evidence.

A technician should not independently declare an incident contained or perform offensive testing unless that responsibility and scope are assigned.

CISA's Secure Our World material is useful for baseline security awareness. Employer security policy and authorized incident procedures remain controlling.

Responsible AI and automation

AI and automation can assist with:

- log summarization;

- drafting ticket notes/runbooks;
- explaining protocol behavior;
- generating lab examples;
- proposing troubleshooting hypotheses;
- repetitive data formatting;
- non-production configuration examples.

Controls:

- use only approved systems and data classes;
- never place credentials, tokens, keys or protected configurations in unapproved public AI services;
- verify every command/configuration before use;
- preserve source/log traceability;
- test appropriately before production;
- do not allow autonomous production changes without approved authority;
- check AI conclusions against observed evidence;
- escalate unexplained or unsafe recommendations.

NIST's AI Risk Management Framework and Generative AI Profile provide voluntary risk-management guidance. They do not replace employer network, security or change governance.

Accessibility and inclusive support

Support processes should be usable by people with disabilities. Useful practices include:

- keyboard-accessible support tools where possible;
- readable headings and structured runbooks;
- meaningful labels;
- sufficient contrast;
- status indicators that do not rely only on color;
- text alternatives for diagrams/screenshots where appropriate;
- communication alternatives when voice-only support is not suitable;
- accessible electronic documents.

Automated accessibility checks do not constitute full legal accessibility certification.

Current U.S. preparation profile

O*NET places **15-1231.00** in **Job Zone Four — Considerable Preparation Needed**. It states that many occupations in this zone require a four-year bachelor's degree, though some do not, and that considerable

related skill/experience and vocational/on-the-job preparation may be needed.

This does not mean every network-support technician job requires a bachelor's degree. Employers vary widely. Relevant pathways can include:

- IT support experience plus networking practice;
- community college or technical education;
- network administration programs;
- certifications where employers value them;
- lab/portfolio evidence;
- related apprenticeship pathways;
- military or employer-provided technical training.

O*NET lists approved example apprenticeship titles **Cloud Support Specialist** and **Junior Cloud Engineer (Nof)**. An approved title does not guarantee a local opening.

United States wages and outlook

BLS 2025 national wage data surfaced through O*NET for 15-1231.00:

Percentile	Annual	Hourly
10th	\$47,120	\$22.65
25th	\$58,240	\$28.00
Median	\$76,220	\$36.64
75th	\$98,750	\$47.48
90th	\$127,780	\$61.43

2024-2034 projections:

- employment 2024: **152,700**;
- projected employment 2034: **155,500**;
- projected growth: **2%**, slower than average;
- projected **annual openings: 9,600**.

Annual openings include growth and replacement openings and should not be interpreted as a guaranteed number of jobs for any individual.

Current non-government related-title context

Indeed reported **\$26.30/hour** average base salary for **Network Technician** in the United States, with a displayed **\$17.46-\$39.60/hour** range, based on approximately **2.1k salaries** from job postings in the prior 36 months, updated **August 3, 2026**.

This is a non-government related-title estimate and may represent a more junior/broader population than *ONET-SOC 15-1231.00*. *Do not substitute it for the official BLS/ONET wage series.*

Current U.S. employer-posting technology signals

O*NET Hot Technologies, based on U.S. postings during 2025, includes:

- Microsoft Office **13%**;
- Microsoft Active Directory **13%**;
- ServiceNow **9%**;
- Linux **7%**;
- Apple macOS **6%**;
- Microsoft Windows Server **6%**;
- Microsoft Outlook **6%**;
- Microsoft Excel **6%**;
- Microsoft Windows **5%**;
- BGP **3%**;
- Microsoft Azure **3%**;
- PowerShell **2%**;
- SQL **1%**;
- Python **1%**;
- Splunk Enterprise **1%**.

These are market signals, not a required checklist for every job.

Canada pathway

Canada Job Bank identifies **NOC 22220 — Computer network and web technicians**. Network technicians establish, operate, maintain and coordinate LANs/WANs and related hardware/software and monitor network connectivity/performance.

Typical requirements currently state:

- completion of a college or other program in computer science, network administration, web technology or a related field is usually required;
- software-vendor certification/training may be required by some employers;
- **registration with a regulatory body is required in Saskatchewan.**

Do not describe the occupation as uniformly unregulated across Canada.

Current national wages, updated November 19, 2025:

- low: **C\$21.00/hour**;
- median: **C\$36.00/hour**;
- high: **C\$55.00/hour**.

Job prospects vary by province/territory. Use the live Job Bank regional outlook for the location being considered.

Colombia pathway

CUOC 35130 — Técnicos en redes y tecnologías de la información

This is a direct occupational match. Relevant titles include:

- Técnico de apoyo de red;
- Técnico de redes y sistemas informáticos;
- Técnico de sistemas en red;
- Técnico de soporte de red informática;
- Técnico en mantenimiento de red informática;
- Técnico en redes de computadores;
- Técnico especialista en infraestructura tecnológica.

Official functions include implementing/operating/troubleshooting data networks, installing network software and operating systems, backup/recovery, configuring interconnection devices, security diagnostics, data-center monitoring, infrastructure maintenance, user support and documentation.

OCUPACOL displays historical/derived salary information but explicitly warns that the data lack statistical representativeness. This guide therefore does **not** use that range as a representative current Colombian national wage benchmark.

SENA — Instalación de redes de computadores

SENA Betowa currently lists **Instalación de redes de computadores** as:

- **Técnico**;
- **2,208 hours**;
- titulada training;
- physical data-network and local wireless-network implementation among the competencies.

Live cohorts, locations and registration dates vary. Verify current availability before planning around a specific intake.

SENA — Gestión de redes de datos

SENA Betowa also lists **Gestión de redes de datos** as:

- **Tecnólogo;**
- **3,984 hours;**
- titulada training;
- structured cabling, data centers, wired/wireless networks and network security among the program themes.

This deeper path can support advancement but is not mandatory for every entry-level network-support position.

Broader Latin America pathway

Training systems vary by country. ILO/Cinterfor's country-level vocational-training resources can help identify national institutions. Verify current program status, admission, modality, cost and credential recognition directly with the provider.

Free-first and affordable learning strategy

Before paying for expensive training:

1. learn networking fundamentals from reputable free/low-cost material;
2. build an isolated home lab or use authorized cloud/lab environments;
3. practice addressing, VLAN, routing, DNS, DHCP, wireless and troubleshooting;
4. document incidents and changes as if working in production;
5. learn one ticketing/monitoring workflow conceptually;
6. then decide whether a vendor certification or formal program matches target employers.

For U.S. readers, CareerOneStop's WIOA locator and training finder can help identify eligible training. Eligibility and funding are determined locally and are not guaranteed.

Ethical portfolio projects

Use only devices, networks, tenants and data you own or have explicit authorization to use.

Portfolio ideas:

- small routed/VLAN lab with diagram and addressing plan;
- DNS/DHCP troubleshooting case;
- wireless fault-isolation scenario;
- simulated branch-to-cloud connectivity design;
- monitoring dashboard using synthetic lab data;
- configuration backup/rollback exercise;
- incident ticket showing evidence, hypothesis, change and validation;

- accessible network runbook;
- automation script that parses synthetic logs without making production changes.

Never scan public systems or a third party's network to create portfolio evidence.

Resume evidence

Strong resume bullets describe outcomes and scope, for example:

- reduced repeat network incidents by improving root-cause documentation;
- restored site connectivity after isolating a VLAN, DHCP or routing fault;
- maintained configuration backups and validated rollback procedures;
- supported routers, switches, wireless and VPN services under change control;
- improved alert/ticket quality through better monitoring evidence.

Use only facts you can support. Do not invent vendor experience, uptime metrics, certifications or security authority.

Interview preparation

Be ready to explain:

- how you troubleshoot when one user cannot connect;
- how your approach changes when an entire site is down;
- difference between DNS and DHCP;
- what a default gateway does;
- what a VLAN is;
- how you would investigate high latency or packet loss;
- why change control and rollback matter;
- what you would do if you saw evidence of compromise;
- how you protect credentials during remote support;
- how you document a resolved incident.

A good answer explains reasoning, safety boundaries and validation—not just commands.

Questions to ask an employer

Ask about:

- network size and major technologies;
- shift/on-call expectations;

- field versus remote work;
- change authority and approval process;
- monitoring/ticketing tools;
- incident escalation;
- documentation standards;
- training/certification support;
- progression to administrator/engineer/cloud/security roles;
- physical requirements and travel;
- accessibility/accommodation process.

First 30 days in a new network-support role

Priorities:

1. learn topology, sites and critical services;
2. learn ticket severity and escalation rules;
3. learn authorized-access boundaries;
4. understand backup/change/rollback procedures;
5. learn monitoring and logging sources;
6. review common incidents and known errors;
7. understand carrier/vendor contacts;
8. verify security and incident-reporting procedures;
9. improve documentation as you learn;
10. do not make undocumented production changes to "prove" capability.

90-day progression plan

Aim to be able to:

- troubleshoot common connectivity issues systematically;
- explain major network dependencies;
- perform assigned changes safely;
- identify when a problem is network versus application/endpoint/service;
- use monitoring/log evidence effectively;
- maintain useful runbooks;
- escalate security concerns correctly;
- identify the next skill path: network administration, cloud, systems, automation or security.

Pre-publication career checklist

Before applying broadly, confirm that you can discuss:

- IPv4/subnet/default gateway;

- DNS and DHCP;
- VLAN/switching basics;
- routing basics;
- wired and wireless troubleshooting;
- VPN/remote connectivity;
- monitoring/logs;
- change/rollback;
- ticket documentation;
- least privilege and security escalation;
- responsible AI boundaries.

Questions before buying training

Ask the provider:

- What occupation/job titles is this program designed for?
- What hands-on network labs are included?
- Are current routing, switching, wireless and cloud-network concepts covered?
- What equipment/software access is included?
- Is certification exam cost included or separate?
- What is the total cost including fees/materials?
- Are outcomes independently verifiable?
- Is funding available and what are the eligibility rules?
- What accessibility accommodations are available?
- Is the credential recognized by target employers?

Do not rely on guaranteed-job or guaranteed-income claims.

Controlled sources

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